



Sec: SC-60

Qualitative Analysis

Date: 04-07-2020

SINGLE ANSWER TYPE

- Sodium nitroprusside solution is used to detect which of the following ions?
a) $S_2O_3^{2-}$ b) Al^{3+} c) Cu^{2+} d) S^{2-}
- Which of the following conditions is not suitable for the brown ring test of NO_2^- ?
a) $FeSO_4$ added must be freshly prepared.
b) H_2SO_4 added should be concentrated.
c) Acetic acid may be used as an alternative acid.
d) Shaking or warming is not allowed.
- On additions of dil. H_2SO_4 to the BaC_2O_4 solution, the gas(es) liberated is/(are)
a) CO only b) CO_2 only. c) $CO + CO_2$ d) None of these
- The NO_3^- ion can be confirmed by
a) addition of conc. H_2SO_4 in the presence of copper turnings.
b) addition of conc. $NaOH$ + Devarda's alloy.
c) the brown ring test only.
d) All of these.
- The metal that does not give the borax bead test is
A) Cr B) Ni C) Pb D) Mn
- $NaCl$, $NaBr$, NaI mixture on adding conc. H_2SO_4 gives respectively
A) HCl , Br_2 , I_2 B) HCl , HBr , HI C) Cl_2 , Br_2 , I_2 D) None of these
- Sodium nitroprusside gives violet colour with sulphide ion. The violet coloured compound is
A) $Na_2[Fe(NO)(CNS)_5]$ B) $H_3[Fe(CN)_6]$
C) $Na_4[Fe(NOS)(CN)_5]$ D) $H_2[Fe(NO)(CN)_5]$
- When Br^- ion is reacted with conc. H_2SO_4 , the gas evolved is
A) Br_2 B) HBr C) HBr & Br_2 D) HBr , Br_2 & SO_2
- A metal salt solution gives a yellow precipitate with silver nitrate. The precipitate dissolves in dilute nitric acid as well as in dilute ammonia solution. The solution contains
A) Bromide ions B) Iodide ions
C) Phosphate ions D) Chromate ions
- Which salt would give SO_2 with hot and dil. H_2SO_4 and also decolourize Br_2 water?
A) Na_2SO_3 B) $NaHSO_4$ C) Na_2SO_4 D) Na_2S
- $[X] + H_2SO_4 \longrightarrow [Y]$, a colourless gas
 $[Y] + K_2Cr_2O_7 + H_2SO_4 \longrightarrow$ green solution. $[X]$ and $[Y]$ are
A) SO_3^{2-} , SO_2 B) Cl^- , HCl C) S^{2-} , H_2S D) CO_3^{2-} , CO_2
- (T) imparts violet colour $\xrightarrow{\text{Compd. (U) + conc. } H_2SO_4}$ (V) Red gas $\xrightarrow{NaOH + AgNO_3}$ (W) red ppt
 $\xrightarrow{NH_3 \text{ soln}}$ (X) (W) Red ppt. $\xrightarrow{\text{dil. } HCl}$ (Y) white ppt.
(U) $\xrightarrow[\Delta]{NaOH}$ (Z) gas (gives white fumes with HCl) sublimes on heating.

Identify (T) to (Z).

- A) $T = \text{KMnO}_4$, $U = \text{HCl}$, $V = \text{Cl}_2$, $W = \text{HgI}_2$, $X = \text{Hg}(\text{NH}_2)\text{NO}_3$, $Y = \text{HgCl}_2$, $Z = \text{N}_2$
 B) $T = \text{K}_2\text{Cr}_2\text{O}_7$, $U = \text{NH}_4\text{Cl}$, $V = \text{CrO}_2\text{Cl}_2$, $W = \text{Ag}_2\text{CrO}_4$, $X = [\text{Ag}(\text{NH}_3)_2]^+$, $Y = \text{AgCl}$, $Z = \text{NH}_3$
 C) $T = \text{K}_2\text{CrO}_4$, $U = \text{KCl}$, $V = \text{CrO}_2\text{Cl}_2$, $W = \text{HgI}_2$, $X = \text{Na}_2\text{CrO}_4$, $Y = \text{BaCO}_3$, $Z = \text{NH}_4\text{Cl}$
 D) $T = \text{K}_2\text{MnO}_4$, $U = \text{NaCl}$, $V = \text{CrO}_3$, $W = \text{AgNO}_2$, $X = (\text{NH}_4)_2\text{CrO}_4$, $Y = \text{CaCO}_3$, $Z = \text{SO}_2$

13. $\text{AgNO}_3 \xrightarrow{\Delta} (\text{W}) + (\text{X}) + \text{O}_2$
 $(\text{X}) + \text{H}_2\text{O} \xrightarrow{\Delta} \text{HNO}_2 + \text{HNO}_3$
 $(\text{W}) + \text{HNO}_3 \xrightarrow{\Delta} \text{Y} + \text{NO} + \text{H}_2\text{O}$
 $(\text{Y}) + \text{Na}_2\text{S}_2\text{O}_3 (\text{excess}) \xrightarrow{\Delta} (\text{Z}) + \text{NaNO}_3$

Identify (W) to (Z).

- | | | | |
|-------------------------------------|---------------------------------|----------------------------|---|
| A) $\text{W} = \text{Ag}$ | $\text{X} = \text{N}_2\text{O}$ | $\text{Y} = \text{AgNO}_3$ | $\text{Z} = \text{Na}_2[\text{Ag}(\text{S}_2\text{O}_3)_2]$ |
| B) $\text{W} = \text{Ag}_2\text{O}$ | $\text{X} = \text{NO}$ | $\text{Y} = \text{AgNO}_3$ | $\text{Z} = \text{Na}_3[\text{Ag}(\text{S}_2\text{O}_3)_2]$ |
| C) $\text{W} = \text{Ag}$ | $\text{X} = \text{NO}_2$ | $\text{Y} = \text{AgNO}_3$ | $\text{Z} = \text{Na}_3[\text{Ag}(\text{S}_2\text{O}_3)_2]$ |
| D) $\text{W} = \text{Ag}_2\text{O}$ | $\text{X} = \text{N}_2$ | $\text{Y} = \text{AgNO}_3$ | $\text{Z} = \text{Na}[\text{Ag}(\text{S}_2\text{O}_3)_2]$ |

14. An aqueous solution of a gas X undergoes the following reactions.
 (i) It turns red litmus blue.
 (ii) When an excess of the solution is added to a copper sulphate solution, a deep blue solution is obtained.
 (iii) On the addition of FeCl_3 solution to a solution of X a brown precipitate, soluble in dilute HNO_3 , is obtained.

Compound X is:

- A) SO_2 B) SO_3 C) NH_3 D) PH_3

15. A bromide does not interfere with the chromyl–chloride test because when a bromide is present,
 A) Br_2 is liberated, which leaves the NaOH solution colourless.
 B) The CrO_2Br_2 does not volatilise as CrO_2Cl_2 does.
 C) CrO_2Br_2 does not react with NaOH
 D) No gaseous compound containing bromine is produced.

MULTIPLE ANSWER TYPE

16. Which of the following acid radicals will produce H_2S gas on action of $[\text{Zn} + \text{dil. H}_2\text{SO}_4]$?
 a) S^{2-} b) SO_4^{2-} c) SO_3^{2-} d) HSO_4^-
17. Which of the following acid radicals produce white precipitate on addition of AgNO_3 ?
 a) $\text{C}_2\text{O}_4^{2-}$ b) NO_2^- c) Cl^- d) HCO_2^-
18. To identify I_2 produced in a particular reaction, the suitable reagents are
 a) excess NaOH solution. b) presence of CCl_4 .
 c) excess KI solution. d) presence of starch

MATCHING TYPE

19. Match the radical with the change observed on qualitative analysis.
- | Column –I | Column –II |
|-----------------------|---|
| a) SO_4^{2-} | P) canary yellow precipitate with ammonium molybdate. |
| b) NO_3^- | Q) Brown ring test with acetic acid. |

c) NO_2^- R) Yellow precipitate with $Hg(NO_3)_2$ solution.

d) PO_4^{3-} S) Evolves NH_3 gas with $(Zn + conc. NaOH)$.

20. Match the compound with the change observed on qualitative analysis.

Column-I

Column-II

a) $(NH_4)_2CO_3$

P) Forms brown ring on reaction with $FeSO_4$ and conc. H_2SO_4

b) $Ag_2C_2O_4$

Q) Leaves no residue on heating.

c) $Pb(NO_3)_2$

R) Colourless acidic gas is evolved on heating.

d) $(NH_4)_2Cr_2O_7$

T) Residue is obtained on heating

e) NH_4NO_3

KEY

1	D	2	B	3	D	4	C	5	C	6	A	7	C	8	A
9	C	10	A	11	A	12	B	13	C	14	C	15	A	16	AC
17	ABCD	18	BCD												

19. A-R B-T C-Q,T D-P

20. A-Q,R B-R,T C-R,T D-PQ E-T